

Notes: Lamont & Polk (JFE, 2002)

Does diversification destroy value? Evidence from industry shocks

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1 Big picture

- **Question.** Does corporate diversification itself destroy shareholder value, or is the diversification discount mainly a selection effect caused by low-value firms choosing to diversify?
- **Core identification problem.** Firms choose whether to diversify, acquire, divest, and report segments. A simple correlation between diversification and value cannot distinguish value destruction from endogenous selection.
- **Main idea.** The paper studies changes in the *within-firm diversity of industry investment opportunities*. It separates total changes in diversity into an exogenous component driven by shocks to industry characteristics and an endogenous component driven by firm structure changes.
- **Main empirical result.** Exogenous increases in investment diversity reduce relative firm value. This implies the diversification discount is not only due to selection or endogenous diversification decisions.
- **Mechanism.** The result supports the inefficient internal capital markets hypothesis: diversified firms have difficulty allocating capital across segments with different investment opportunities.
- **Measurement strategy.** The paper avoids the mechanical measurement-error problem of using industry Q both to build the right-hand-side diversity measure and the left-hand-side excess value measure. It measures investment-opportunity diversity using industry investment-to-capital ratios rather than industry Q .
- **Key economic magnitude.** Investment diversity explains more than 40% of the average diversification discount in the sample.
- **Additional evidence.** Direct segment investment evidence shows diversified firms smooth segment capital expenditures more than comparable focused firms, consistent with internal cross-subsidization.
- **Robustness.** The negative exogenous diversity effect survives controls for lagged excess value, lagged diversity, firm and year fixed effects, outlier removal, manufacturing-only samples, four-digit SIC benchmarks, and alternative measures of diversity.
- **Broader takeaway.** Diversification is not always destructive in every conceivable setting, but conditional on already being diversified, exogenous increases in unrelatedness appear to reduce value.

2 Data and measurement

2.1 Sample construction and Compustat segment data

- The sample uses Compustat segment data for 1979–1997, with the main annual-change panel covering 1980–1997.
- The main sample contains 11,974 firm-year observations from 1,987 diversified firms.
- A firm is treated as diversified in year t if it has multiple segments in year $t - 1$. This convention allows the authors to study firms that reduce the number of segments between $t - 1$ and t .
- Firms that move from focused in $t - 1$ to diversified in t are excluded from the change-sample because they have no exogenous diversity change over the prior structure.
- The paper excludes financial firms, firms with sales below \$20 million, firm-years with inconsistencies between segment and firm totals, and cases lacking firm-level equity, debt, investment, capital stock, or sales growth data.
- Segment-level variables used in the main analysis are segment sales, segment assets, and segment SIC codes. Segment capital expenditures are used later for a direct segment-investment test.
- Focused firms are single-segment firms used to construct industry benchmarks at the two-digit SIC level.

Interpretation: The sample design is meant to make diversified firms comparable to portfolios of focused firms. The key empirical variation comes from diversified firms that keep a set of segments but face changing industry-level investment conditions.

2.2 Firm value and excess value

The firm-level market-book ratio is

$$Q_{it} = \frac{\text{market value of common stock} + \text{book assets} - \text{book equity} - \text{deferred taxes}}{\text{book assets}}. \quad (1)$$

For each segment j of firm i , the paper finds focused firms in the same two-digit SIC industry and computes the industry median market-book ratio, Q_{jt}^{IND} . The imputed value ratio of the diversified firm is

$$\bar{Q}_{it} = \sum_{j=1}^{n_i} w_{ijt} Q_{jt}^{IND}, \quad (2)$$

where w_{ijt} is the segment asset weight. Excess value is

$$q_{it} - \bar{q}_{it} = \ln \left(\frac{Q_{it}}{\bar{Q}_{it}} \right). \quad (3)$$

The dependent variable in the main regressions is the annual change in excess value,

$$\Delta(q - \bar{q})_{it} = (q_{it} - \bar{q}_{it}) - (q_{i,t-1} - \bar{q}_{i,t-1}). \quad (4)$$

Interpretation: Excess value measures whether the diversified firm is worth more or less than a portfolio of focused firms operating in the same industries. Using annual changes helps reduce concerns about persistent firm-level omitted variables.

2.3 Industry investment and investment diversity

For each focused firm, investment-to-capital is capital expenditures in year t divided by the prior-year net stock of property, plant, and equipment. For each two-digit industry, the benchmark I_{jt}^{IND} is the median investment-to-capital ratio among focused firms in the segment's industry.

For a diversified firm with n_i segments, investment diversity is the within-firm standard deviation of industry investment benchmarks:

$$\sigma_{it} = sd_j(I_{jt}^{IND}). \quad (5)$$

A focused firm mechanically has diversity equal to zero. A diversified firm with segments in industries that have very different investment rates has high diversity.

Interpretation: This measure is intended to proxy for diversity in investment opportunities without using industry Q , which is already used to construct excess value. That design reduces mechanical measurement error.

2.4 Exogenous and endogenous changes in diversity

Let $structure_t$ denote the firm's set of segments and SIC codes in year t , and let $benchmark_t$ denote industry investment benchmarks in year t . Then total diversity change can be written as

$$\Delta\sigma_t = \sigma(structure_t, benchmark_t) - \sigma(structure_{t-1}, benchmark_{t-1}) \quad (6)$$

$$= \underbrace{[\sigma(structure_t, benchmark_t) - \sigma(structure_{t-1}, benchmark_t)]}_{\Delta\sigma_N} \quad (7)$$

$$+ \underbrace{[\sigma(structure_{t-1}, benchmark_t) - \sigma(structure_{t-1}, benchmark_{t-1})]}_{\Delta\sigma_X}. \quad (8)$$

- $\Delta\sigma_X$ is the **exogenous diversity change**: how diversity would have changed if the firm had kept its old segment structure but industry investment rates changed.
- $\Delta\sigma_N$ is the **endogenous diversity change**: how diversity changes because the firm changes segment structure, number of segments, or reported SIC assignments.
- $\Delta\sigma = \Delta\sigma_X + \Delta\sigma_N$.

Interpretation: The decomposition is the core empirical design. Industry shocks are plausibly outside the control of any single diversified firm, while segment additions, deletions, and reclassifications are firm choices or firm reporting choices.

2.5 Alternative diversity measures

The paper also studies:

- asset-weighted investment diversity σ_1 ;
- weighted standard deviation of industry investment σ_2 ;
- Rajan, Servaes, and Zingales style asset-weighted industry- Q diversity;
- diversity in leverage, sales growth, and cash flow.

Interpretation: Alternative measures test whether the main investment-diversity result is an artifact of equal weighting, of a particular formula, or of unrelatedness more generally.

2.6 Direct segment investment data

For a subset of diversified firms whose segments each have at least \$20 million in sales, the paper uses actual segment capital expenditures divided by segment sales, C/S . It constructs:

- $\sigma_{C/S-Segment}$: within-firm standard deviation of actual segment C/S ;
- $\sigma_{C/S-Focus}$: standard deviation of C/S across matched focused firms.

Interpretation: The inefficient internal capital markets hypothesis predicts that diversified firms smooth investment across segments relative to focused firms. If they do, actual within-firm segment investment dispersion should be lower than the dispersion among matched focused firms.

3 Models and empirical specifications

3.1 Baseline change regression

The core regression is

$$\Delta(q - \bar{q})_{it} = \alpha + \beta \Delta\sigma_{it} + u_{it}. \quad (9)$$

The decomposed version is

$$\Delta(q - \bar{q})_{it} = \alpha + \beta_X \Delta\sigma_{X,it} + \beta_N \Delta\sigma_{N,it} + u_{it}. \quad (10)$$

Interpretation: A negative β_X means that a purely industry-driven increase in investment diversity lowers diversified-firm value. Because $\Delta\sigma_X$ holds prior firm structure fixed, it is the key coefficient for causal interpretation.

3.2 Baseline fixed-effects specification

The preferred specification adds lagged excess value, lagged diversity, firm fixed effects, and year fixed effects:

$$\Delta(q - \bar{q})_{it} = \alpha_i + \lambda_t + \beta_X \Delta\sigma_{X,it} + \beta_N \Delta\sigma_{N,it} + \rho(q - \bar{q})_{i,t-1} + \phi\sigma_{i,t-1} + u_{it}. \quad (11)$$

Interpretation: Lagged excess value controls for mean reversion in relative value. Lagged diversity controls for predictable changes in diversity, because expected industry changes should already be priced. Firm fixed effects absorb time-invariant firm characteristics, and year effects absorb aggregate shocks.

3.3 Measurement-error framework

The paper is concerned that excess value and diversity may share measurement error. A simplified measurement-error model is

$$\Delta\bar{q}^{obs} = \Delta\bar{q}^{true} + e, \quad (12)$$

where error in the imputed benchmark $\bar{q}$ can enter the dependent variable and potentially also the right-hand-side diversity measure. To reduce this concern, the baseline diversity measure uses industry investment rather than industry Q .

Interpretation: The paper’s main defense is design-based: the dependent variable uses industry Q , while the main treatment variable uses industry investment. Table 4 then quantifies whether remaining measurement error can plausibly explain the results.

3.4 Alternative specification checks

The paper estimates versions of the baseline using:

- manufacturing-only firms;
- market-sales ratios instead of market-book ratios;
- coefficient of variation rather than standard deviation;
- Fama-MacBeth annual cross-sectional regressions;
- four-digit SIC industry benchmarks with no five-focused-firm minimum.

Interpretation: These checks ask whether the baseline result depends on sample composition, value metric, diversity functional form, estimation method, or industry classification.

3.5 Direct segment investment smoothing specification

The direct segment-investment test examines whether actual segment investment diversity moves one-for-one with focused-firm benchmark diversity:

$$\Delta\sigma_{C/S-Segment,it} - \Delta\sigma_{C/S-Focus,it} = \alpha + \gamma\Delta\sigma_{C/S-Focus,it} + u_{it}. \quad (13)$$

Interpretation: If diversified firms smooth segment investment relative to focused firms, then when focused-firm investment dispersion rises, actual segment dispersion should rise less. This produces a negative coefficient in the difference regression.

3.6 Other-diversity regressions

For other industry characteristics Z such as leverage, sales growth, and cash flow, the paper estimates

$$\Delta(q - \bar{q})_{it} = \alpha_i + \lambda_t + \beta_Z^X \Delta\sigma_{Z,X,it} + \beta_Z^N \Delta\sigma_{Z,N,it} + \beta_I^X \Delta\sigma_{X,it} + \beta_I^N \Delta\sigma_{N,it} + controls + u_{it}. \quad (14)$$

Interpretation: These regressions test whether the investment-diversity effect is really a proxy for broader unrelatedness, leverage mismatch, demand-growth mismatch, or cash-flow mismatch.

4 Identification strategy

4.1 Step 1: Start with an endogenous-diversification problem

The diversification discount is hard to interpret because firms choose diversification. A negative relation between diversification and value may arise because:

- diversification destroys value;
- low-value firms choose to diversify;
- firms diversify after unobserved negative shocks;
- firms acquire low-value segments or divest high-value segments;
- measurement error mechanically links imputed value and diversity measures.

Interpretation: The paper’s goal is not just to document the discount, but to isolate variation in diversity that is plausibly not chosen by the firm.

4.2 Step 2: Use industry shocks to create exogenous diversity changes

The identifying assumption is that changes in industry investment rates are exogenous from the perspective of an individual diversified firm. Holding the firm’s prior segment structure fixed, industry shocks can make its segments more or less different in investment opportunities.

Interpretation: If a firm with aircraft and electronics segments becomes less diverse because the aircraft industry’s investment rises toward electronics, this variation comes from industry conditions rather than a firm decision to refocus or diversify.

4.3 Step 3: Separate exogenous from endogenous diversity

The decomposition $\Delta\sigma = \Delta\sigma_X + \Delta\sigma_N$ separates:

- changes caused by industry benchmark movements, interpreted as exogenous;
- changes caused by segment structure changes, interpreted as endogenous.

Interpretation: The causal claim rests mainly on $\Delta\sigma_X$, not on $\Delta\sigma_N$. A negative coefficient on $\Delta\sigma_N$ is suggestive but cannot determine whether firm choices caused value destruction or responded to falling value.

4.4 Step 4: Use changes, lagged controls, firm fixed effects, and year effects

The preferred specification uses annual changes in excess value, plus lagged excess value, lagged diversity, firm effects, and year effects.

Interpretation: This design removes much of the persistent cross-sectional heterogeneity behind the usual diversification discount. The coefficient is identified from within-firm changes in industry-driven diversity.

4.5 Step 5: Reduce mechanical measurement-error bias

Measurement error is particularly important because excess value uses industry Q benchmarks. The paper avoids using industry Q to build the main diversity measure and instead uses industry investment. It also directly assesses how much measurement error would be needed to generate the observed coefficient.

Interpretation: This is a key methodological contribution. The authors acknowledge the measurement-error critique and design the baseline measure specifically to avoid “hardwiring” the result.

4.6 Step 6: Check the internal capital markets channel

The main value regressions cannot directly show the exact mechanism. The paper therefore examines actual segment capital expenditures and finds that diversified firms smooth segment investment relative to focused firms.

Interpretation: This evidence links the value result to capital allocation behavior: diversified firms appear to spend less differentially across high- and low-opportunity segments than focused firms do.

4.7 Threats and limits

- Industry shocks may affect diversified firms through channels other than internal capital misallocation.
- Industry investment rates are proxies for investment opportunities, not perfect measures.
- Segment reporting changes may be noisy and sometimes reflect accounting classification rather than real organizational change.
- The design identifies the effect of diversity conditional on already being diversified, not the value effect of randomly assembling diversified firms from focused firms.
- The paper cannot fully distinguish cash-flow value destruction from discount-rate effects.

Interpretation: The evidence is strongest for the claim that exogenous increases in investment-opportunity dispersion reduce relative value inside diversified firms. It is less direct on the complete welfare evaluation of diversification or the precise managerial agency channel.

5 Tables and figures

5.1 Table 1: Structure, value, and diversity for Northrop Grumman Corp, 1996–1997

Read:

- Northrop Grumman had aircraft and electronics segments in 1996 and added an information technology/services segment in 1997.
- Its excess value improved from -0.290 to -0.209 , so $\Delta(q - \bar{q}) = 0.081$.
- Total diversity rose from 0.110 to 0.196 , so $\Delta\sigma = 0.086$.
- Holding the 1996 structure fixed, industry changes alone would have reduced diversity to 0.072 , so $\Delta\sigma_X = -0.038$.
- The endogenous structure change raised diversity by $\Delta\sigma_N = 0.124$.

Interpretation: This table shows the decomposition in a concrete firm. Industry shocks made the old aircraft/electronics mix less diverse, but the addition of a new high-investment segment increased total diversity.

Table 1

Structure, value, and diversity for Northrop Grumman Corp, 1996–1997

Q is the market-book ratio of the firm, calculated as the ratio of the market value of the firm (the market value of common stock plus the book value of total asset minus book equity minus deferred taxes) divided by the total book assets of the firm. Q_{IND} is the median Q among focused firms in the segment's industry. $\bar{Q}$ is the imputed value ratio of the firm, calculated as the weighted average of the Q_{IND} 's for each of the firm's segments, where the weights are the book assets of the segments. I_{IND} is the median investment to capital ratio among focused firms in the segment's industry, where investment to capital is year t capital expenditures divided by year $t - 1$ net capital stock. σ is diversity, the within-firm standard deviation of I_{IND} . Lower case letters indicate natural logs. $\Delta\sigma_X$ is the exogenous change in the standard deviation of I_{IND} , defined as σ (year t structure, year t benchmarks) minus σ (year $t - 1$ structure, year t benchmarks). $\Delta\sigma_N$ is the endogenous change in the standard deviation of I , $\Delta\sigma_N = \Delta\sigma - \Delta\sigma_X$.

	1996		1997		Information technology and services
	Aircraft	Electronics	Aircraft	Electronics	
SIC	37	38	37	38	73
Book assets	2387	5970	2386	5451	559
% total	0.29	0.71	0.28	0.65	0.07
Q_{IND}	1.283	1.872	1.474	1.990	2.357
I_{IND}	0.176	0.332	0.250	0.352	0.629
<i>Firm level variables</i>					
$\bar{Q}$		1.275			1.515
$\bar{Q}$		1.703			1.868
$q - \bar{q}$		-0.290			-0.209
Δq					0.173
$\Delta \bar{q}$					0.092
$\Delta q - \Delta \bar{q}$					0.081
σ		0.110			0.196
σ using 1996 structure and 1997 investment					0.072
$\Delta\sigma$					0.086
$\Delta\sigma_X$					-0.038
$\Delta\sigma_N$					0.124

Table 2

Summary statistics

Statistics for the sample of 11,974 firm-year observations from 1,987 diversified firms. Q is the market-book ratio of the firm, calculated as the ratio of the market value of the firm (the market value of common stock plus the book value of total asset minus book equity minus deferred taxes) divided by the total book assets of the firm. $\bar{Q}$ is the imputed value ratio of the firm using the segment's industry Q 's, where the industry Q 's are median Q 's for single segment firms in the same two-digit industry as the segment, and the weights are the book assets of the segments. σ is diversity, the within-firm standard deviation of industry median investment to capital ratios. $\Delta\sigma_X$ is the exogenous change in the standard deviation of industry investment, defined as σ (year t structure, year t benchmarks) minus σ (year $t - 1$ structure, year t benchmarks). $\Delta\sigma_N$ is the endogenous change in the standard deviation of industry investment, $\Delta\sigma_N = \Delta\sigma - \Delta\sigma_X$. Lagged $q - \bar{q}$ is excess value in year $t - 1$. Lagged σ is the standard deviation of industry investment in year $t - 1$. Lower case letters indicate natural logs. The sample period is 1980–1997.

	Mean	Median	Std	Min	Max
Q	1.338	1.149	0.660	0.350	12.998
$\bar{Q}$	1.299	1.242	0.293	0.711	5.054
$q - \bar{q}$	-0.028	-0.061	0.351	-1.863	1.848
Δq	0.012	0.016	0.220	-2.179	2.083
$\Delta \bar{q}$	0.022	0.032	0.126	-0.592	1.060
$\Delta q - \Delta \bar{q}$	-0.010	-0.006	0.214	-2.239	2.042
σ	0.049	0.037	0.051	0.000	0.652
$\Delta\sigma$	-0.002	0.000	0.045	-0.618	0.525
$\Delta\sigma_X$	0.001	0.000	0.040	-0.618	0.525
$\Delta\sigma_N$	-0.003	0.000	0.026	-0.474	0.194

Correlation matrix

	Δq	$\Delta \bar{q}$	$\Delta q - \Delta \bar{q}$	$\Delta\sigma$	$\Delta\sigma_X$	$\Delta\sigma_N$	Lagged $q - \bar{q}$
Δq	0.332						
$\Delta q - \Delta \bar{q}$	0.831	-0.249					
$\Delta\sigma$	-0.055	-0.007	-0.052				
$\Delta\sigma_X$	-0.043	-0.014	-0.036	0.822			
$\Delta\sigma_N$	-0.029	0.009	-0.036	0.472	-0.113		
Lagged $q - \bar{q}$	-0.269	0.053	-0.307	0.010	0.004	0.011	
Lagged σ	0.022	-0.012	0.029	-0.464	-0.403	-0.185	-0.038

Economic message: The identification strategy is not just “diversified vs. focused.” It asks whether diversity changes because industry opportunities move or because the firm changes its structure.

5.2 Table 2: Summary statistics

Read:

- The sample has 11,974 firm-years from 1,987 diversified firms.
- Mean and median excess value are negative: mean $q - \bar{q} = -0.028$ and median -0.061 .
- Mean diversity is 0.049, while mean total diversity change is near zero.
- The standard deviation of $\Delta\sigma_X$ is 0.040 and the standard deviation of total $\Delta\sigma$ is 0.045, so most diversity-change variation comes from industry shocks.
- $\Delta\sigma_X$ and $\Delta\sigma$ are highly correlated, while $\Delta\sigma_N$ is less central to total diversity changes.

Interpretation: The sample exhibits a diversification discount and substantial variation in exogenous diversity shocks. The data therefore have enough within-firm, industry-driven movement to test the paper's question.

Economic message: The table motivates the design: exogenous shocks explain much of the annual movement in diversity, so industry shocks can be used to study value effects.

5.3 Table 3: Basic regression of change in excess value on change in diversity

Read:

- Column (1) shows total diversity change is negatively related to excess value changes: $\Delta\sigma$ coefficient -0.250 .

- Column (2) uses exogenous diversity changes only: $\Delta\sigma_X = -0.193$.
- Column (3) includes both components: $\Delta\sigma_X = -0.217$ and $\Delta\sigma_N = -0.338$.
- Column (4), the baseline with lagged variables and firm/year fixed effects, gives $\Delta\sigma_X = -0.302$ and $\Delta\sigma_N = -0.324$.
- Lagged excess value has a large negative coefficient, consistent with mean reversion.

Interpretation: The key result is the negative coefficient on $\Delta\sigma_X$. Since this component is driven by industry conditions holding prior firm structure fixed, the result is evidence that diversity itself can reduce value.

Economic message: A one standard deviation increase in diversity lowers excess value economically meaningfully. Investment diversity alone can explain more than 40% of the average diversification discount.

Table 3

Basic regression of change in excess value on change in diversity

The dependent variable is $\Delta q - \Delta \bar{q}$, the change in excess value. $\Delta\sigma$ is the change in the standard deviation of industry investment. $\Delta\sigma_X$ is the exogenous change in the standard deviation of industry investment, caused only by changes in industry characteristics. $\Delta\sigma_N$ is the endogenous change in the standard deviation of industry investment, caused only by change in corporate structure including number and SIC code of segments. Lagged $q - \bar{q}$ is excess value in year $t - 1$. Lagged σ is the standard deviation of industry investment in year $t - 1$. The number of observations is 11,974; the sample period is 1980–1997. “Year and firm dummies” indicate separate intercepts for each of 18 years and each of the 1987 firms. The standard errors, in parentheses, are calculated allowing for both heteroskedasticity and for the residuals to be correlated within each of the 18 years, 1980–1997.

	(1)	(2)	(3)	(4)
$\Delta\sigma$	-0.250 (0.052)			
$\Delta\sigma_X$		-0.193 (0.066)	-0.217 (0.062)	-0.302 (0.069)
$\Delta\sigma_N$			-0.338 (0.120)	-0.324 (0.124)
Lagged $q - \bar{q}$				-0.470 (0.027)
Lagged σ				-0.197 (0.095)
Constant	-0.010 (0.007)	-0.010 (0.007)	-0.010 (0.007)	
R^2	0.003	0.001	0.003	0.366
Year and firm dummies	N	N	N	Y

Table 4

Robustness to measurement error and outliers

In regressions (1)–(3), the dependent variable is $\Delta \bar{q}$, the change in the natural log of $\bar{Q}$. $\bar{Q}$ is the imputed value ratio, the weighted average of the segment's industry Q 's. $\Delta\sigma$ is the change in the standard deviation of industry investment ratios. $\Delta\sigma_X$ is the exogenous change in the standard deviation of industry investment ratios, caused only by changes in industry characteristics. $\Delta\sigma_N$ is the endogenous change in the standard deviation of industry investment ratios, caused only by change in corporate structure. Lagged $q - \bar{q}$ is excess value in year $t - 1$. Lagged σ is the standard deviation of industry investment in year $t - 1$. In regressions (4)–(6), the dependent variable is $\Delta q - \Delta \bar{q}$, the change in excess value. “No outliers” discards observations of $\Delta q - \Delta \bar{q}$, $\Delta\sigma_X$, and $\Delta\sigma_N$, that are in the top or bottom 0.5% of their distributions. “Year and firm dummies” indicate separate intercepts for each year and firm. The number of observations is 11,974 (except for column (6), which is 11,624); the sample period is 1980–1997. The standard errors, in parentheses, are calculated allowing for both heteroskedasticity and for the residuals to be correlated within each year.

	$\Delta \bar{q}$ on left-hand side			$\Delta q - \Delta \bar{q}$ on left-hand side		
	(1)	(2)	(3)	(4)	(5)	No outliers (6)
$\Delta\sigma$	-0.020 (0.117)					
$\Delta\sigma_X$		-0.045 (0.146)	0.024 (0.107)	-0.212 (0.072)	-0.289 (0.071)	-0.277 (0.095)
$\Delta\sigma_N$			-0.003 (0.065)		-0.326 (0.103)	-0.337 (0.056)
Lagged $q - \bar{q}$			0.049 (0.012)		-0.444 (0.024)	-0.399 (0.028)
Lagged σ			-0.139 (0.060)		-0.272 (0.085)	-0.239 (0.088)
$\Delta \bar{q}$				-0.423 (0.039)		
Constant	0.022 (0.021)	0.022 (0.021)		-0.000 (0.008)	0.411	0.340
R^2	0.000	0.000	0.555	0.064		
Year and firm dummies	N	N	Y	N	Y	Y

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5.4 Table 4: Robustness to measurement error and outliers

Read:

- Columns (1)–(3) use the change in imputed value $\Delta \bar{q}$ as the dependent variable to assess whether measurement error can mechanically generate the baseline result.
- The estimated mechanical effects are small relative to the baseline negative coefficient on $\Delta\sigma_X$.
- Columns (4)–(5) add $\Delta \bar{q}$ to the excess-value regression; the exogenous diversity effect remains negative.
- Column (6) excludes extreme observations; the coefficients on $\Delta\sigma_X$ and $\Delta\sigma_N$ remain negative.

Interpretation: The table directly addresses the critique that the result is an artifact of common benchmark construction. The main specification is comparatively immune to this problem because it uses industry investment for diversity rather than industry Q .

Economic message: The diversification-destroys-value conclusion is not driven by a mechanical relationship between the imputed value benchmark and the diversity measure.

5.5 Table 5: Alternative specifications

Read:

- The manufacturing-only sample yields a strongly negative coefficient on $\Delta\sigma_X$.

- Using market-sales excess value rather than market-book excess value still gives negative effects.
- Measuring diversity with a coefficient of variation rather than a standard deviation gives negative exogenous and endogenous effects.
- Fama-MacBeth estimation gives a negative exogenous effect.
- Four-digit SIC benchmarks with no minimum focused-firm rule produce weaker $\Delta\sigma_X$ estimates but still negative endogenous-diversity estimates.

Interpretation: These tests show that the baseline is not sensitive to one narrow definition of value, one diversity formula, or one estimation procedure.

Economic message: The relation between investment-opportunity diversity and lower value appears robust across reasonable measurement choices.

Table 5

Alternative specifications

The dependent variable in regressions (1) and (3) through (5) is $\Delta q - \Delta \bar{q}$, the change in excess value. $\Delta\sigma_X$ is the exogenous change in the standard deviation of industry investment, caused only by changes in industry characteristics. $\Delta\sigma_N$ is the endogenous change in the standard deviation of industry investment, caused only by change in corporate structure. Lagged variables are as of year $t-1$. Column (1) restricts the sample to diversified firms with all segments in manufacturing industries, defined as two-digit SIC codes 20 through 39. In column (2), the dependent variable is $\Delta m - \Delta \bar{m}$, the change in excess value measured using market-sales ratios, defined as the log ratio of the total market value to total firm sales. In column (3), diversity in industry investment is defined as the coefficient of variation of industry investment. CV is the coefficient of variation for industry investment, defined as the standard deviation of industry investment divided by the mean of industry investment. Lagged variables are as of year $t-1$. ΔCV_X is the exogenous change in the coefficient of variation of industry investment, defined as CV (year t structure, year t benchmarks) minus CV (year $t-1$ structure, year t benchmarks). ΔCV_N is the endogenous change in the standard deviation of industry investment. $\Delta CV_X = \Delta CV - \Delta CV_N$. Column (4) uses Fama-Macbeth estimation. Column (5) repeats the baseline specification using four-digit SIC codes instead of two-digit SIC codes in calculating all variables, and allowing the industry benchmarks to come from as few as one firm (instead of requiring at least five firms). The sample period is 1980-1997. "Year and firm dummies" indicate separate intercepts for each year and firm. The standard errors in columns (1)-(3), and (5), in parentheses, are calculated allowing for both heteroskedasticity and for the residuals to be correlated within each year.

	Manufacturing only	Dependent variable is $\Delta m - \Delta \bar{m}$	Diversity measured with coefficient of variation	Fama-Macbeth estimation	Four-digit SIC codes, no five firm minimum
	(1)	(2)	(3)	(4)	(5)
$\Delta\sigma_X$	-0.461 (0.081)	-0.397 (0.118)		-0.213 (0.066)	-0.020 (0.003)
$\Delta\sigma_N$	-0.274 (0.225)	-0.935 (0.305)		-0.217 (0.118)	-0.243 (0.093)
Lagged $q - \bar{q}$	-0.457 (0.037)		-0.470 (0.027)	-0.175 (0.021)	-0.571 (0.031)
Lagged σ	-0.223 (0.069)	-0.340 (0.135)		0.006 (0.048)	-0.009 (0.022)
Lagged $m - \bar{m}$		-0.480 (0.023)			
ΔCV_X			-0.064 (0.017)		
ΔCV_N			-0.063 (0.027)		
Lagged CV			-0.049 (0.019)		
Lagged $Q - \bar{Q}$					
R^2	0.375	0.365	0.365		0.433
Firm and year dummies	Y	Y	Y	N	Y
Observations	6438	11974	11974	11974	6119

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Table 6

Alternative measures of investment diversity

The dependent variable is $\Delta q - \Delta \bar{q}$, the change in excess value. σ_1 is the standard deviation of asset-weighted industry investment. σ_2 is the weighted standard deviation of industry investment.

$$\sigma_1 = \sqrt{\frac{1}{n-1} \sum_{j=1}^n \left(w_j I_{INDj} - \left[\frac{1}{n} \sum_{i=1}^n w_i I_{INDi} \right] \right)^2} \quad \text{and}$$

$$\sigma_2 = \sqrt{\frac{n}{n-1} \sum_{j=1}^n w_j \left(I_{INDj} - \left[\sum_{i=1}^n w_i I_{INDi} \right] \right)^2}$$

where w_j is the fraction of the firm's assets in segment j , I_{INDj} is industry investment for segment j , and the firm has n segments. $\Delta\sigma_1X$ is the exogenous change in the standard deviation of asset-weighted industry investment and $\Delta\sigma_1N$ is the endogenous change. $\Delta\sigma_2X$ is the exogenous change in the weighted standard deviation of industry investment and $\Delta\sigma_2N$ is the endogenous change. $\Delta\sigma_X$ is the exogenous change in the standard deviation of industry investment and $\Delta\sigma_N$ is the endogenous change. Lagged variables are as of year $t-1$. The number of observations is 11,974. All regressions include separate intercepts for each year and firm. The standard errors, in parentheses, are calculated allowing for both heteroskedasticity and for the residuals to be correlated within each year.

	(1)	(2)	(3)	(4)
$\Delta\sigma_1X$	-0.439 (0.168)	-0.377 (0.163)		
$\Delta\sigma_1N$	-0.102 (0.066)	-0.019 (0.062)		
$\Delta\sigma_2X$			-0.348 (0.085)	-0.140 (0.219)
$\Delta\sigma_2N$			-0.461 (0.164)	-0.508 (0.300)
$\Delta\sigma_X$		-0.258 (0.067)		-0.194 (0.166)
$\Delta\sigma_N$		-0.332 (0.127)		0.040 (0.209)
Lagged $q - \bar{q}$	-0.471 (0.028)	-0.472 (0.027)	-0.470 (0.027)	-0.470 (0.027)
Lagged σ_1	-0.244 (0.080)	-0.218 (0.077)		
Lagged σ_2			-0.195 (0.105)	0.139 (0.290)
Lagged σ		-0.174 (0.095)		-0.311 (0.261)
Joint significance of both exogenous changes (p -value)		0.000		0.000
R^2	0.366	0.368	0.366	0.367

5.6 Table 6: Alternative measures of investment diversity

Read:

- The table uses asset-weighted diversity σ_1 and weighted standard deviation σ_2 .
- Exogenous changes in σ_1 and σ_2 are generally negative and often significant.
- When the baseline $\Delta\sigma_X$ is added, the combined exogenous diversity terms remain jointly significant.
- The result therefore does not depend on treating each segment equally.

Interpretation: The main baseline measure ignores segment weights, but Table 6 shows that weighting segments by assets does not overturn the result.

Economic message: Value destruction is linked to differences in segment investment opportunities, not merely to a quirk of equal-weight standard deviations.

5.7 Table 7: Direct evidence from segment capital expenditures

Read:

- Actual diversified-firm segment investment dispersion has mean 0.058, while matched focused-firm dispersion has mean 0.064.
- The difference is negative and statistically significant.
- The median difference is also negative and significant.
- In the regression, the coefficient on $\Delta\sigma_{C/S-Focus}$ is -0.978 , close to the one-for-one smoothing implication.

Interpretation: Diversified firms' actual segment investment is less dispersed than the investment of comparable focused firms. This is direct evidence that internal capital allocation smooths investment across segments.

Economic message: The table supports the mechanism behind the value regressions: when segments face different opportunities, diversified firms do not allocate investment as sharply as focused firms do.

Table 7

Direct evidence from segment capital expenditures

Summary statistics and regression results for actual segment investment and investment of matching focused firms. C/S is the ratio of capital expenditures to sales for the individual segment or focused firm. $\sigma_{C/S-Segment}$ is the standard deviation of segment C/S for each diversified firm, and $\sigma_{C/S-Focus}$ is the standard deviation of matching firm C/S for each diversified firm. The sample is limited to diversified firms for which all segments have at least \$20 million in sales. For means, the p -value is from a T -test, and for medians, the p -value is from a Wilcoxon signed-rank test. The number of observations is 7,514 for summary statistics, and 7,002 for regression results. For the regression results, the standard errors, in parentheses, are calculated allowing for both heteroskedasticity and for the residuals to be correlated within each year.

Summary statistics			
	$\sigma_{C/S-Segment}$	$\sigma_{C/S-Focus}$	Difference
Mean	0.058	0.064	-0.006 p -value = 0.011
Median	0.022	0.031	-0.009 p -value = 0.000
Regression results			
Dependent variable: $\Delta\sigma_{C/S-Segment} - \Delta\sigma_{C/S-Focus}$			
Constant	-0.003	(0.002)	
$\Delta\sigma_{C/S-Focus}$	-0.978	(0.028)	
R^2	0.429		

Table 8

Diversity in asset-weighted Q

The dependent variable is $\Delta Q - \Delta \bar{Q}$. RSZ is diversity in asset-weighted Q , defined as the standard deviation of asset-weighted Q , divided by $\bar{Q}_{EW} = (1/n) \sum_j \bar{Q}_{IND,j}$, the equal-weighted average of $\bar{Q}_{IND,j}$.

$$RSZ = \frac{1}{\bar{Q}_{EW}} \sqrt{\frac{\sum_{j=1}^n (w_j \bar{Q}_{IND,j} - [(1/n) \sum_{j=1}^n w_j \bar{Q}_{IND,j}])^2}{n-1}}$$

where w_j is the fraction of the firm's assets in segment j , and $\bar{Q}_{IND,j}$ is the benchmark Q for segment j 's industry. ΔRSZ_X is the exogenous change in RSZ , caused only by changes in industry characteristics. ΔRSZ_N is the endogenous change in RSZ , caused only by change in corporate structure including number and SIC code of segments and weights. $\Delta \ln(Sales)$ is the log change in total firm sales. $\Delta \sigma_N$ is the exogenous change in the standard deviation of industry investment ratios, caused only by changes in industry characteristics. $\Delta \sigma_N$ is the endogenous change in the standard deviation of industry investment ratios, caused only by change in corporate structure. The number of observations is 11,974. All regressions include separate intercepts for each year and firm. The standard errors, in parentheses, are calculated allowing for both heteroskedasticity and for the residuals to be correlated within each year.

	$\Delta Q - \Delta \bar{Q}$ on left-hand side			$\Delta \bar{Q}$ on left-hand side	
	(1)	(2)	(3)	(4)	(5)
$ARSZ$	-0.116 (0.047)				
ΔRSZ_X		-0.664 (0.156)	-0.459 (0.138)	0.906 (0.127)	1.290 (0.064)
ΔRSZ_N		-0.076 (0.052)			0.036 (0.010)
$\Delta 1/\bar{Q}_{EW}$	0.947 (0.140)	0.985 (0.147)			-1.667 (0.080)
$\Delta \ln(Sales)$	-0.082 (0.044)	-0.084 (0.044)			0.003 (0.007)
Constant			-0.015 (0.012)	0.028 (0.028)	
R^2	0.223	0.225	0.001	0.030	0.861
Firm and year dummies	Y	Y	N	N	Y

5.8 Table 8: Diversity in asset-weighted Q

Read:

- The table studies the Rajan, Servaes, and Zingales measure of asset-weighted industry- Q diversity.
- ΔRSZ is negatively related to changes in excess value.
- The exogenous component ΔRSZ_X is also negative in value regressions.
- However, because industry Q enters both the treatment and outcome construction, measurement-error concerns are more severe.
- When $\Delta \bar{Q}$ is the dependent variable, the table shows mechanical relationships that caution against overinterpreting Q -based diversity regressions.

Interpretation: The table confirms that Q -based diversity measures show similar negative patterns, but it also illustrates why the paper prefers investment-based diversity.

Economic message: Industry- Q diversity is informative but potentially contaminated. Industry investment is the cleaner baseline because it is less directly tied to excess value construction.

5.9 Table 9: Diversity in leverage, cash flow, and sales growth

Read:

- Leverage diversity has negative endogenous effects but little evidence of a causal exogenous effect.
- Sales-growth diversity does not robustly explain excess value once investment diversity is included.
- Cash-flow diversity has negative exogenous and endogenous effects.
- Including cash-flow diversity reduces but does not eliminate the negative effect of exogenous investment diversity.

Interpretation: Other kinds of unrelatedness do not fully subsume investment diversity. Cash-flow diversity has its own negative effect, consistent with a broader cross-subsidization or unrelatedness channel.

Economic message: Investment diversity is a distinct source of value destruction, while cash-flow diversity may reflect an additional internal-capital-market problem.

Table 9
Diversity in leverage, cash flow, and sales growth
The dependent variable is $\Delta q - \Delta q$, the change in excess value. Leverage is the book value of debt divided by the sum of market value of equity and the book value of debt. Sales growth is net sales in year t divided by net sales in year $t - 1$. Cash flow is income before extraordinary items plus depreciation in year t divided by capital stock in year $t - 1$. $\Delta \sigma_X$ is the exogenous change in the standard deviation of industry investment ratios, caused only by changes in industry characteristics. $\Delta \sigma_X$ is the endogenous change in the standard deviation of industry investment ratios, caused only by change in corporate structure. $\Delta \sigma Z_X$ is the exogenous change in the standard deviation of industry variable Z , caused only by changes in industry characteristics. $\Delta \sigma Z_X$ is the endogenous change in the standard deviation of industry variable Z , caused only by change in corporate structure. Lagged variables are as of year $t - 1$. All regressions include separate intercepts for each year and firm. The number of observations is 11,974; the sample period is 1980–1997. The standard errors, in parentheses, are calculated allowing for both heteroskedasticity and for the residuals to be correlated within each year.

	Leverage: $Z = \frac{D_t}{E_t + D_t}$		Sales growth: $Z = \frac{S_t}{S_{t-1}}$		Cash flow: $Z = \frac{CF_t}{K_{t-1}}$	
$\Delta \sigma Z_X$	-0.122 (0.081)	-0.095 (0.085)	-0.015 (0.101)	0.086 (0.112)	-0.151 (0.043)	-0.126 (0.040)
$\Delta \sigma Z_N$	-0.264 (0.070)	-0.184 (0.081)	-0.447 (0.126)	-0.263 (0.145)	-0.241 (0.051)	-0.229 (0.064)
$\Delta \sigma_X$		-0.251 (0.076)		-0.296 (0.082)		-0.210 (0.069)
$\Delta \sigma_N$		-0.187 (0.143)		-0.210 (0.154)		-0.045 (0.139)
Lagged $q - q$	-0.470 (0.028)	-0.471 (0.027)	-0.469 (0.028)	-0.470 (0.027)	-0.470 (0.028)	-0.470 (0.027)
Lagged σZ	-0.183 (0.034)	-0.132 (0.040)	-0.209 (0.114)	-0.097 (0.124)	-0.161 (0.038)	-0.145 (0.038)
Lagged σ		-0.130 (0.100)		-0.153 (0.099)		-0.053 (0.094)
Joint significance of $\Delta \sigma Z_X$ and $\Delta \sigma Z_N$ (p -value)	0.000	0.074	0.002	0.136	0.000	0.000
R^2	0.366	0.367	0.365	0.367	0.367	0.368

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6 Interpretive synthesis

6.1 What the paper adds to the diversification-discount literature

The paper moves the diversification-discount literature from cross-sectional correlation toward causal inference. Prior evidence showed that diversified firms are valued less than comparable focused portfolios. Lamont and Polk ask whether exogenous changes in within-firm diversity reduce value. The answer is yes: when industry shocks make a diversified firm's segments more different in investment opportunities, relative value falls.

6.2 Why industry investment diversity is the key object

Investment diversity is economically meaningful because the internal capital markets hypothesis is about allocation across high- and low-opportunity segments. If segments' opportunities are very different, a weak internal capital market can more easily misallocate funds. Using industry investment rather than industry Q also helps avoid a mechanical relation between right-hand-side diversity and left-hand-side excess value.

6.3 How to interpret endogenous diversity

Endogenous diversity changes are also negatively related to value, but they are harder to interpret. They may reflect value-destroying acquisitions, value-preserving responses to negative shocks, divestitures of valuable segments, or noisy segment reporting changes. The paper therefore emphasizes exogenous diversity.

6.4 Mechanism versus reduced form

The main regressions identify a reduced-form value effect of exogenous diversity changes. The segment investment evidence then supports the mechanism: diversified firms smooth investment more than focused firms do. Together, these results point toward inefficient internal capital markets, but they do not fully identify the exact agency model causing misallocation.

7 Short summary

- The paper asks whether diversification destroys value or merely reflects endogenous choices by low-value firms.
- The data cover diversified firms in Compustat segment files from 1980–1997.
- Excess value is the log ratio of firm Q to an imputed focused-firm portfolio Q .
- Investment diversity is the within-firm standard deviation of industry investment-to-capital benchmarks across the firm’s segments.
- Total diversity change is decomposed into exogenous industry-driven change and endogenous structure-driven change.
- Exogenous changes in investment diversity have a negative and significant effect on changes in excess value.
- The baseline coefficient on $\Delta\sigma_X$ is about -0.302 with lagged controls and firm/year fixed effects.
- Measurement error is unlikely to explain the baseline result because diversity is measured with industry investment rather than industry Q .
- Alternative samples, value measures, diversity formulas, and estimation methods support the result.
- Direct segment investment evidence shows diversified firms smooth investment across segments relative to matched focused firms.
- Other diversity measures do not subsume investment diversity; cash-flow diversity also has an independent negative effect.
- The conclusion is that diversification destroys value, conditional on already being diversified, and the evidence supports inefficient internal capital markets.

8 Conclusion

8.1 Core conclusion

Lamont and Polk provide evidence that corporate diversification destroys value. The key result is that exogenous increases in the diversity of industry investment opportunities reduce relative firm value. This means the diversification discount is not solely a selection effect from low-value firms choosing to diversify.

8.2 Economic intuition

Diversity in investment opportunities creates a capital allocation problem. If one segment has strong opportunities and another has weak opportunities, an efficient internal capital market should direct funds toward the strong segment. If internal politics, fairness norms, managerial incentives, or headquarters constraints smooth investment across divisions, then diversity becomes costly. The larger the dispersion in opportunities, the larger the potential misallocation.

8.3 Identification takeaway

The paper's key methodological idea is to use industry shocks to separate exogenous diversity changes from endogenous restructuring. This provides a general empirical strategy for studying organizational choices when firms select into the treatment.

8.4 Limitations

The paper does not show that randomly combining focused firms always destroys value. It studies the effect of exogenous diversity changes among already-diversified firms. It also does not fully separate cash-flow effects from discount-rate effects or identify the exact agency friction behind inefficient allocation.

8.5 Final takeaway

The paper strengthens the causal interpretation of the diversification discount. Diversification appears to reduce value when it exposes firms to greater dispersion in segment investment opportunities, and the evidence is consistent with inefficient internal capital markets.